

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 04

**BOOLEAN THEOREMS AND
GATES** - 1

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Recap of Previous Lecture

Logic gates



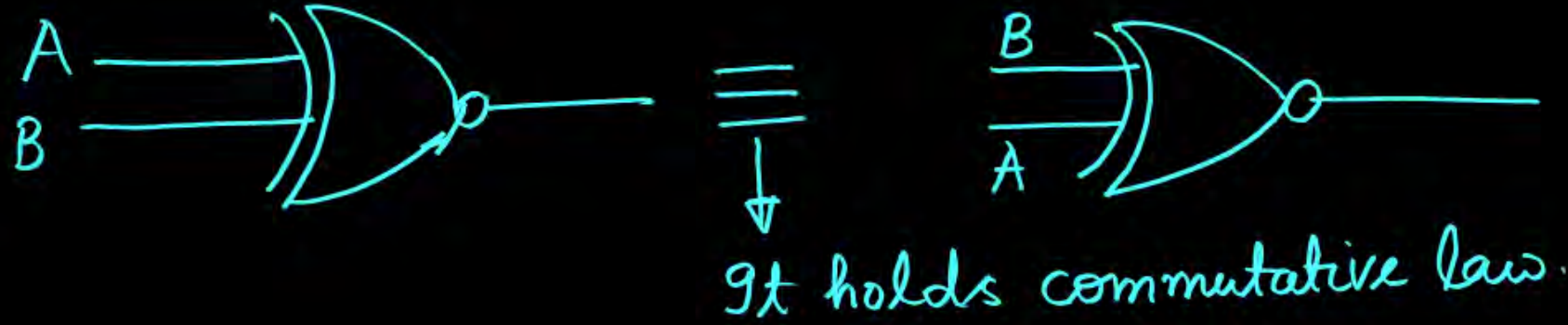


Topics to be Covered

XNOR gate cont.

Question Discussion.

- Commutative Law :



$$A \oplus B = B \oplus A$$

- Associative Law : It holds associative law $\rightarrow$ defined to calculate multi i/p XNOR operation.

$$1 \odot 0 \odot 1 \odot 0 = 1$$

$$A \odot B \odot C \rightarrow A \odot B \rightarrow (A \odot B) \odot C$$

$$A \odot C \rightarrow (A \odot C) \odot B$$

$$B \odot C \rightarrow (B \odot C) \odot A$$

$$A \odot B \odot C \odot D \rightarrow (A \odot B) \odot (C \odot D)$$

$$\rightarrow A \odot B \rightarrow (A \odot B) \odot C \rightarrow [(A \odot B) \odot C] \odot D$$

- Properties of XNOR gate :

- $A \odot A = 1 = \bar{A} \cdot \bar{A} + A \cdot A$
 $\bar{A} \odot \bar{A} = 1$
 $AB \odot AB = 1$

- $A \odot \bar{A} = 0 = \bar{A} \cdot A + A \cdot \bar{A}$
 $AB \odot \overline{AB} = 0$

- $A \odot 1 = A = \bar{A} \cdot 0 + A \cdot 1$
 $\bar{A} \odot 1 = \bar{A}$
 $AB \odot 1 = \bar{AB}$

- $A \odot 0 = \bar{A} = \bar{A} \cdot 1 + A \cdot 0$
 $\bar{A} \odot 0 = \bar{\bar{A}}$
 $AB \odot 0 = \overline{AB}$

- Exchange properties of XNOR gate :

$$\text{If } A \odot B = C$$

$$\text{then } \longrightarrow A \odot C = B$$

$$\& \longrightarrow B \odot C = A$$

$$\Rightarrow A \odot C = A \odot A \odot B = 1 \odot B = B$$

$$\Rightarrow B \odot C = B \odot A \odot B = 1 \odot A = A$$

- $A \odot A \odot A \odot A \dots n \text{ times}$ [n - represents no. of A]

then o/p = 1 for n -even

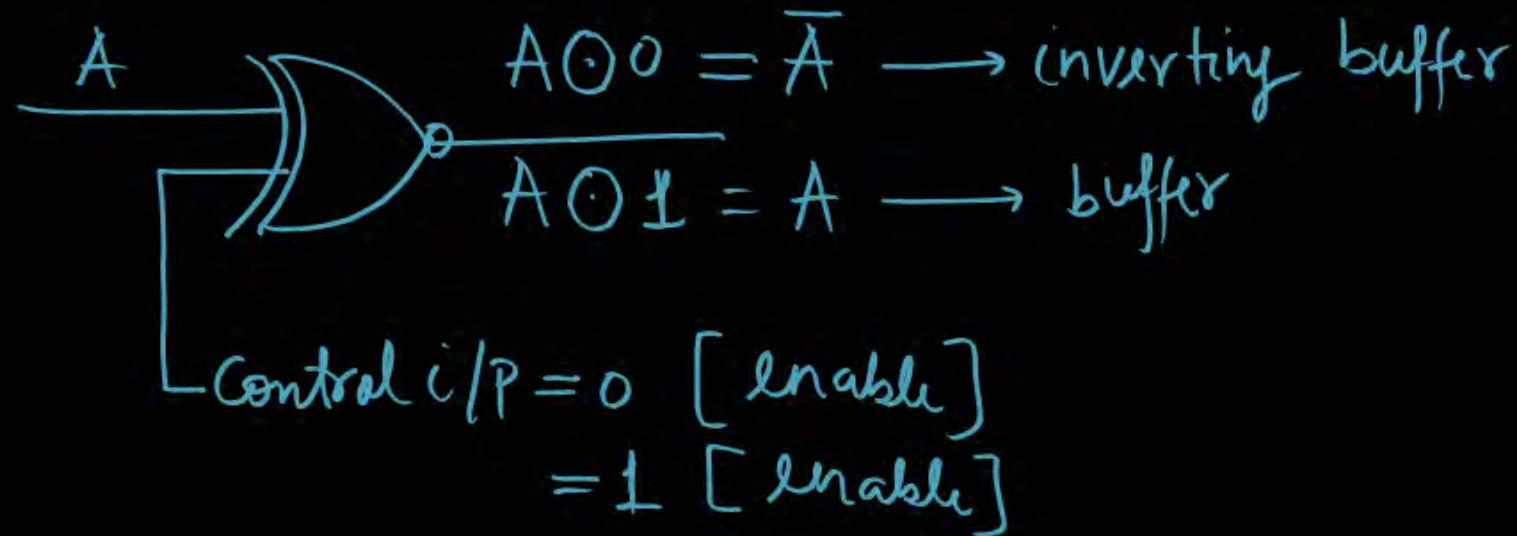
= A for n -odd

$$A \odot A = 1$$

$$A \odot A \odot A \odot A = 1 \odot 1 = 1$$

$$A \odot A \odot A = 1 \odot A = A$$

- Buffer and inverting buffer using XNOR :



[Combined IMP Properties of XOR and XNOR]

- $$\begin{aligned}\overline{A} \oplus B &= \overline{\overline{A}} \cdot B + \overline{A} \cdot \overline{B} = A \odot B \\ A \oplus \overline{B} &= \overline{A} \cdot \overline{B} + A \cdot \overline{\overline{B}} = A \odot B \\ \overline{A} \oplus \overline{B} &= A \cdot \overline{B} + \overline{A} \cdot B = A \oplus B\end{aligned}$$

- $$\begin{aligned}\overline{A} \odot B &= A \overline{B} + \overline{A} \cdot B = A \oplus B \\ A \odot \overline{B} &= \overline{A} B + A \cdot \overline{B} = A \oplus B \\ \overline{A} \odot \overline{B} &= A \cdot B + \overline{A} \cdot \overline{B} = A \odot B\end{aligned}$$

- $(A \oplus B) \oplus (\bar{A} \oplus C) \oplus (B \oplus C) \oplus (\bar{A} \oplus \bar{C})$

$$A \oplus \bar{A} \oplus \bar{A} = A$$

$$B \oplus B = 0$$

$$C \oplus C \oplus \bar{C} = 0 \oplus \bar{C} = \bar{C}$$

$$A \oplus 0 \oplus \bar{C} = A \oplus \bar{C} = A \odot C$$

- $(A \oplus \bar{B}) \oplus (C \odot B) \oplus (A \odot C) \oplus (B \oplus C) \rightarrow B \oplus \bar{C} \oplus 0$
 $= B \oplus \bar{C} = B \odot C$

$$A \oplus \bar{B} \oplus \bar{C} \oplus B \oplus \bar{A} \oplus C \oplus B \oplus C$$

$$= 1 \oplus \bar{B} \oplus \bar{C}$$

$$= B \oplus \bar{C}$$

$$= B \odot C$$



- $A \oplus B \oplus C = \overline{A \odot B \odot C} \rightarrow$ true or false $A \oplus B \oplus C = A \odot B \odot C$

	A	B	C	$y_1 = A \oplus B \oplus C$	$y_2 = A \odot B \odot C$
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	0
4	1	0	0	1	1
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

$$y_1[A, B, C] = \sum(1, 2, 4, 7) = A \oplus B \oplus C = A \odot B \odot C$$

$$A \oplus B \oplus C = [A \oplus B] \oplus C = [\bar{A}B + A\bar{B}] \oplus C$$

$$= (\bar{A}\bar{B} + AB)C + (\bar{A}B + A\bar{B})\bar{C}$$

$$= \bar{A}\bar{B}C + ABC + \bar{A}B\bar{C} + A\bar{B}\bar{C} = \Sigma(1, 2, 4, 7)$$

$\begin{matrix} 001 & 111 & 010 & 100 \end{matrix}$

$$\bullet A \odot B \odot C = [A \odot B] \odot C = (\bar{A}\bar{B} + AB) \odot C$$

$$= (\bar{A}B + A\bar{B})\bar{C} + (\bar{A}\bar{B} + AB)C$$

$$= \bar{A}B\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}C + ABC$$

$\begin{matrix} 010 & 100 & 001 & 111 \end{matrix}$

$$= \Sigma(1, 2, 4, 7)$$

- $1 \oplus 0 \oplus 1 \oplus 0 = 0$

$$1 \oplus 1 \oplus 1 \oplus 0 = 1$$

$$1 \oplus 0 \oplus 0 \oplus 0 = 1$$

$$1 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 1$$

$$1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 = 0$$

$$1 \odot 0 \odot 1 \odot 0 = 1$$

$$1 \odot 1 \odot 1 \odot 0 = 0$$

$$1 \odot 0 \odot 0 \odot 0 = 0$$

$$1 \odot 1 \odot 0 \odot 1 \odot 0 = 1$$

$$1 \odot 0 \odot 1 \odot 0 \odot 0 = 0$$

From above we conclude that :

- XOR gate search for $\rightarrow$ always odd no. of 1's $\rightarrow$ i.e. when there are odd no. of 1's $\rightarrow$ o/p is '1' otherwise '0'.

- XNOR gate search for $\rightarrow$ odd no. of 1's for odd no. of i/p lines (odd no. of variables)
 $\rightarrow$ even no. of 1's for even no. of i/p lines (even no. of variables)

• That's why $\rightarrow$

$$A \oplus B \oplus C = A \odot B \odot C$$

$$A \oplus B \oplus C \oplus D = \overline{A \odot B \odot C \odot D}$$

$$A \oplus B \oplus C \oplus D \oplus E = A \odot B \odot C \odot D \odot E$$

- Compliment property of XOR and XNOR GATE :

$$\overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B} = A \odot B$$

$$\overline{A \odot B} = \bar{A} \odot B = A \odot \bar{B} = A \oplus B$$

$$\bar{A} \oplus \bar{B} = A \odot B = A \oplus B$$

- $$\overline{A \oplus B \oplus C} = \bar{A} \oplus B \oplus C = A \oplus \bar{B} \oplus C = A \oplus B \oplus \bar{C}$$

$$= A \odot B \oplus C = A \oplus B \odot C$$

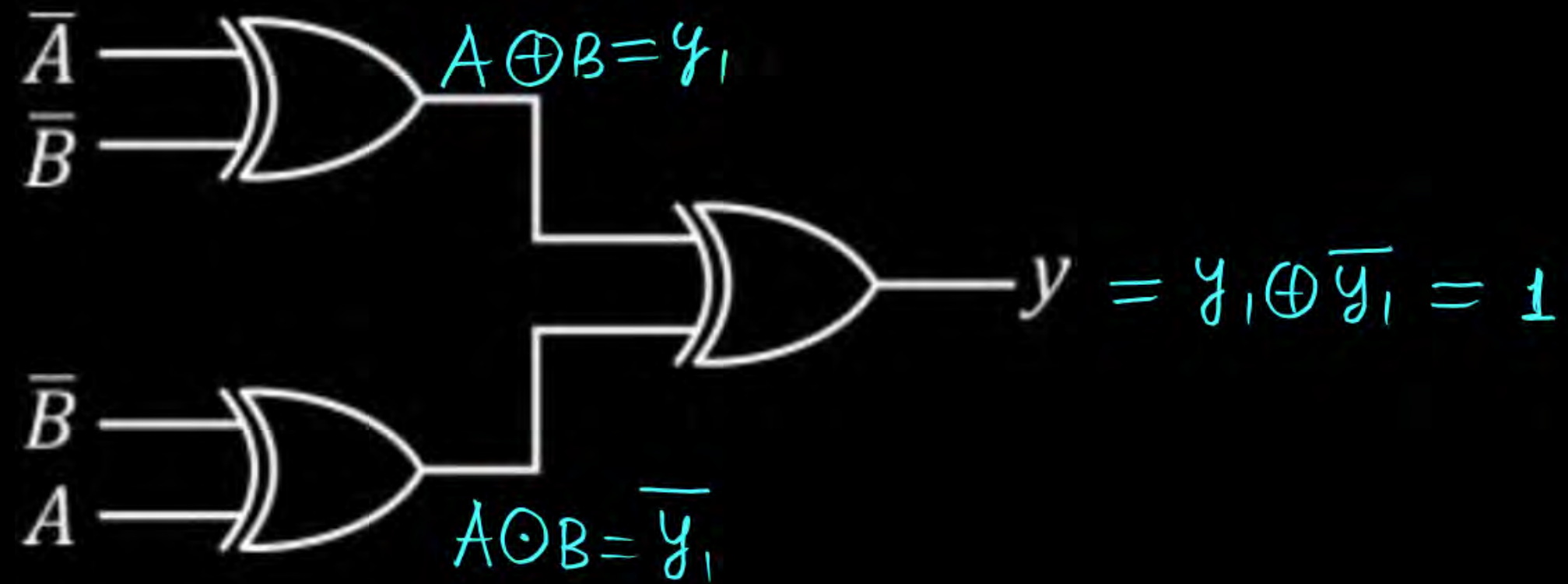
$$\overline{\overline{A \oplus B \oplus C}} = \overline{A \odot B \oplus C} = \bar{A} \odot B \oplus C = A \odot \bar{B} \oplus C = A \odot B \oplus \bar{C}$$

$$= A \oplus B \oplus C = A \odot B \odot C$$

$$\begin{aligned}
 & \bullet \quad \overline{\overline{A \odot B} \oplus \overline{C} \oplus \overline{D}} \\
 &= \overline{A \odot B \oplus C \oplus D} \\
 &= A \oplus B \oplus C \oplus D \\
 &= \overbrace{A \odot B \odot C}^{\uparrow \quad \uparrow} \oplus D
 \end{aligned}
 \qquad
 \begin{aligned}
 &= \overline{\overline{A \oplus B \oplus C} \oplus \overline{D}} \\
 &= \overline{A \oplus B \oplus C \oplus \overline{D}} \\
 &= A \oplus B \oplus C \oplus D
 \end{aligned}$$

$$\begin{aligned}
 & \bullet \quad \overline{A \oplus \overline{B} \odot \overline{C} \oplus \overline{D}} \\
 &= A \oplus B \odot C \oplus D \\
 &= \overline{A \oplus B \oplus C \oplus D}
 \end{aligned}$$

[Question]



Output y is

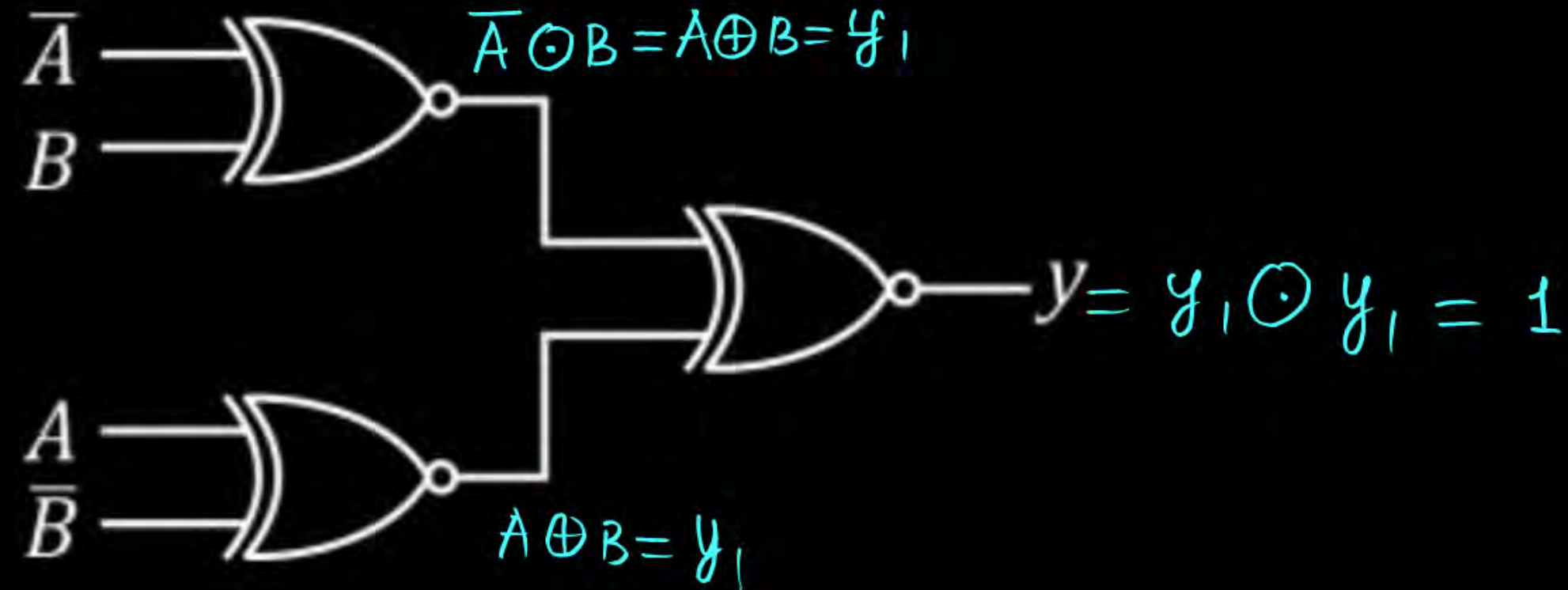
(a) $A \odot B$

(b) $A \oplus B$

☒ (c) Always '1'

(d) Always '0'

[Question]



Output y is

(a) $A \oplus B$

(b) $A \odot B$

(c) ✓ Always '1'

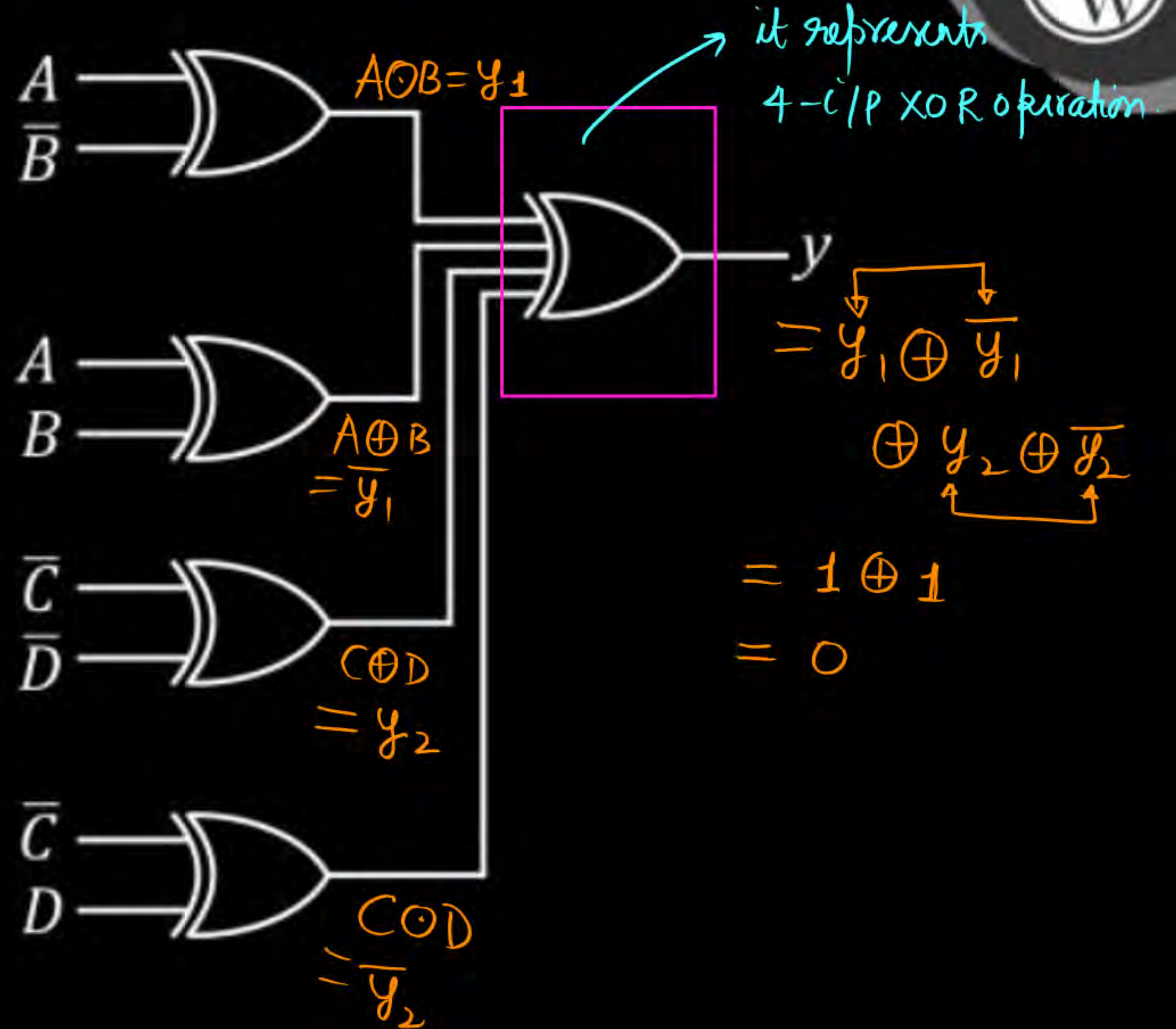
(d) Always '0'

[Question]



Output y is

- (a) $A \odot B \oplus C \oplus D$
- (b) $A \oplus B \oplus C \oplus D$
- (c) Always '0'
- (d) $A \odot B \odot C \odot D$



[Question]

$y = \overline{\overline{A \oplus B \oplus C}}$ will be equal to $= \overline{A \oplus B \oplus C}$

(a) ✓ $A \odot B \oplus C$ ✓

(b) $y = \overline{\overline{A} \oplus \overline{B \odot C}} = \overline{A \oplus B \odot C}$

(c) $y = \overline{A \oplus B \oplus C}$

(d) ✗ None of these

$$= A \oplus B \oplus C$$

$$= A \odot B \oplus C$$

$$= A \oplus B \oplus C$$

[Question]

$y = A \oplus [A + B]$ equal to

(a) $A \oplus B$ ✗

(b) $y = A \odot B$

(c) ✓ $\bar{A}B$

(d) $A\bar{B}$

$$A \oplus [A + B] \neq (A \oplus A) + A \oplus B = (A \oplus B)$$

$$\cdot \underline{A \oplus (A + B)} = \bar{A} (A + B) + A \cdot \overline{(A + B)} = \bar{A}B + A \cdot \bar{A} \cdot \bar{B} = \bar{A}B$$

Note: XOR & XNOR operation does not hold distributive law over 'OR' & 'AND' operation.

[Question]

$y = A \oplus B \oplus AB$ equal to

(a) AB

(c) $\overline{AB}$

(b) $(A + B)$

(d) $\overline{A + B}$

✓
 $A \oplus B \oplus AB = (A + B)$
 $A \oplus B \oplus (A + B) = A \cdot B$

• $A \oplus B \oplus AB = A \oplus [B \oplus AB] = A \oplus [\overline{B} \cdot AB + B \cdot (\overline{A} + \overline{B})]$
 $A \oplus \overline{A}B \neq A \odot AB$
 $A \oplus \overline{AB} = A \odot AB$
 $= \underline{A} \oplus \underline{\overline{A}B} = \overline{A} \cdot \overline{A}B + A \cdot (A + \overline{B})$
 $= \overline{A}B + A$
 $= A + \overline{A}B = (A + B)$

Thank you

GW
Soldiers !

